

Name/*Naam*:

Student #:

Question 1:**Vraag 1:**

2.1 Determine the indefinite integral of $\int (\cos \theta - \sin 2\theta) d\theta$
Bereken die onbepaalde integraal van

2.2 Determine the definite integral of $\int_0^{\pi/2} (\cos \theta - \sin 2\theta) d\theta$
Bereken die bepaalde integraal van

Answer the following questions below and on reverse side of page./Antwoord die volgende onderaan en op agter kant van hierdie bladsy.

Question 2 / Vraag 2*HRW Chapter 3 Problem 39***Question 3 / Vraag 3***HRW Chapter 18 Problem 15*

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Question 1:

Vraag 1:

- 1.1 Determine the indefinite integral of $\int (\cos \theta - \sin 2\theta) d\theta$
1.2 Bereken die onbepaalde integraal van

- 2.1 Determine the definite integral of $\int_{\pi/2}^{3\pi/2} (\cos \theta - \sin 2\theta) d\theta$
Bereken die bepaalde integraal van

Answer the following questions below and on reverse side of page./*Antwoord die volgende onderaan en op agter kant van hierdie bladsy.*

Question 2 / Vraag 2

HRW Chapter 3 Problem 40 a and c

Question 3 / Vraag 3

HRW Chapter 18 Problem 14 a and b

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Question 1:**Vraag 1:**

2.2 Determine the indefinite integral of $\int (\cos \theta - \sin 2\theta) d\theta$
Bereken die onbepaalde integraal van

2.3 Determine the definite integral of $\int_{\pi/2}^{\pi} (\cos \theta - \sin 2\theta) d\theta$
Bereken die bepaalde integraal van

Answer the following questions below and on reverse side of page./Antwoord die volgende onderaan en op agter kant van hierdie bladsy.

Question 2 / Vraag 2*HRW Chapter 3 Problem 39***Question 3 / Vraag 3***HRW Chapter 18 Problem 14 c*

Question 1:

Vraag 1:

1.1 Determine the indefinite integral of

Bereken die onbepaalde integraal van

$$\int (\cos \theta - \sin 2\theta) d\theta = \sin \theta + \frac{1}{2} \cos 2\theta + c$$

1.2 Determine the definite integral of

Bereken die bepaalde integraal van

$$\begin{aligned} \int_0^{\pi/2} (\cos \theta - \sin 2\theta) d\theta &= [\sin \theta + \frac{1}{2} \cos 2\theta]_0^{\pi/2} \\ &= [1 + (-\frac{1}{2})] - [0 + \frac{1}{2}] = 0 \end{aligned}$$

$$\begin{aligned} \int_{\pi/2}^{3\pi/2} (\cos \theta - \sin 2\theta) d\theta &= [\sin \theta + \frac{1}{2} \cos 2\theta]_{\pi/2}^{3\pi/2} \\ &= [-1 - \frac{1}{2}] - [1 - \frac{1}{2}] = -2 \end{aligned}$$

$$\begin{aligned} \int_{\pi/2}^{\pi} (\cos \theta - \sin 2\theta) d\theta &= [\sin \theta + \frac{1}{2} \cos 2\theta]_{\pi/2}^{\pi} \\ &= [0 + \frac{1}{2}] - [1 - \frac{1}{2}] = 0 \end{aligned}$$

Chapter 3 Vectors

Q 39

From the definition of the dot product between $\vec{A}$ and $\vec{B}$, $\vec{A} \cdot \vec{B} = AB \cos \theta$, we have

$$\begin{aligned} \cos \theta &= \frac{\vec{A} \cdot \vec{B}}{AB} = \frac{14.0}{(6.00)(7.00)} = 0.333 \\ \theta &= \cos^{-1}(0.333) = 70.5^\circ \end{aligned}$$

Q40 a and c

The displacement vectors can be written as (in meters)

$$\begin{aligned} \vec{d}_1 &= (4.80 \text{ m})(\cos 63^\circ \hat{j} + \sin 63^\circ \hat{k}) = (2.18 \text{ m})\hat{j} + (4.28 \text{ m})\hat{k} \\ \vec{d}_2 &= (1.40 \text{ m})(\cos 30^\circ \hat{i} + \sin 30^\circ \hat{k}) = (1.21 \text{ m})\hat{i} + (0.70 \text{ m})\hat{k} . \end{aligned}$$

(a) The dot product of $\vec{d}_1$ and $\vec{d}_2$ is

$$\vec{d}_1 \cdot \vec{d}_2 = (2.18\hat{j} + 4.28\hat{k}) \cdot (1.21\hat{i} + 0.70\hat{k}) = (4.28\hat{k}) \cdot (0.70\hat{k}) = 3.00 \text{ m}^2.$$

(c) The magnitudes of $\vec{d}_1$ and $\vec{d}_2$ are

$$d_1 = \sqrt{(2.18 \text{ m})^2 + (4.28 \text{ m})^2} = 4.80 \text{ m}$$

$$d_2 = \sqrt{(1.21 \text{ m})^2 + (0.70 \text{ m})^2} = 1.40 \text{ m}.$$

Thus, the angle between the two vectors is

$$\theta = \cos^{-1} \left(\frac{\vec{d}_1 \cdot \vec{d}_2}{d_1 d_2} \right) = \cos^{-1} \left(\frac{3.00 \text{ m}^2}{(4.80 \text{ m})(1.40 \text{ m})} \right) = 63.5^\circ.$$

Chapter 18

Q 15

After the change in temperature the diameter of the steel rod is $D_s = D_{s0} + \alpha_s D_{s0} \Delta T$ and the diameter of the brass ring is $D_b = D_{b0} + \alpha_b D_{b0} \Delta T$, where D_{s0} and D_{b0} are the original diameters, α_s and α_b are the coefficients of linear expansion, and ΔT is the change in temperature. The rod just fits through the ring if $D_s = D_b$. This means

$$D_{s0} + \alpha_s D_{s0} \Delta T = D_{b0} + \alpha_b D_{b0} \Delta T.$$

Therefore,

$$\begin{aligned} \Delta T &= \frac{D_{s0} - D_{b0}}{\alpha_b D_{b0} - \alpha_s D_{s0}} = \frac{3.000 \text{ cm} - 2.988 \text{ cm}}{(19.00 \times 10^{-6} / \text{C}^\circ)(2.988 \text{ cm}) - (11.00 \times 10^{-6} / \text{C}^\circ)(3.000 \text{ cm})} \\ &= 505.0^\circ\text{C}. \end{aligned}$$

The temperature is $T = (25.00^\circ\text{C} + 505.0^\circ\text{C}) = 530.0^\circ\text{C}$.

Q14

(a) Since $A = \pi D^2/4$, we have the differential $dA = 2(\pi D/4)dD$. Dividing the latter relation by the former, we obtain $dA/A = 2 dD/D$. In terms of Δ 's, this reads

$$\frac{\Delta A}{A} = 2 \frac{\Delta D}{D} \quad \text{for} \quad \frac{\Delta D}{D} \ll 1.$$

We can think of the factor of 2 as being due to the fact that area is a two-dimensional quantity. Therefore, the area increases by $2(0.20\%) = 0.40\%$.

(b) Assuming that all dimensions are allowed to freely expand, then the thickness increases by 0.20%.

(c) The volume (a three-dimensional quantity) increases by $3(0.20\%) = 0.60\%$.